SEC-B-2

(Information Security)

Full Marks : 80

Group - A

1. Answer ***any ten*** questions : 2×10

   (a) How information security different from cybersecurity?

   (b) How is Encryption different from Hashing?

   (c) What is a firewall and how does it help in securing a network?

   (d) Find 2^{16} mod 5 using Fermat's theorem.

   (e) What is Steganography?

   (f) What is the difference between a virus and a worm in the context of computer security?

   (g) What is digital certificate?

   (h) Highlight the main differences between active and passive attack.

   (i) What is symmetric key encryption? Name two commonly used symmetric algorithms.

   (j) What is a fabrication attack?

   (k) What do you understand by Brute-force attack?

   (l) What is Trojan horse?

   (m) How does 'denial of service' prevent the management of communication facilities?

   (n) What is the purpose of log management in computer security?

   (o) What is Encapsulating Security Payload?

Group - B

2. Answer ***any four*** questions : 5×4

   (a) Discuss briefly about SSL attacks.

   (b) Explain the features and working of S/MIME in securing e-mail communication.

   (c) What is Message digest? Why is it used?

   (d) Explain Euler's theorem with suitable example.

   (e) Write a short note on Pretty Good Privacy (PGP). How does it ensure e-mail security?

   (f) What is IPsec? Explain its components and how it secures IP communications.

Group - C

Answer *any four* questions.

3. (a) Write short notes on Web and Wireless security.

   (b) What do you mean by Secure Electronic Transaction (SET)? Explain with appropriate illustration. 5+5

4. (a) Explain Chinese Remainder theorem.

   (b) Write RSA algorithm with a suitable example. 5+5

5. (a) Explain the Diffie-Hellman key exchange algorithm with an example.

   (b) What is the CIA triad in information security? Briefly explain each component. 5+5

6. (a) What do you mean by Galois field?

   (b) What are the principal elements of Private-key and Public-key Cryptosystem? 5+5

7. Define groups, rings and fields in the context of number theory. How are they useful in cryptography? (2+2+2)+4

8. (a) Explain the basic principle of the AES algorithm.

   (b) Write a short note on Transport Layer Security (TLS). 5+5